Proposal Report
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1. DESCRIPTION OF ASCEND

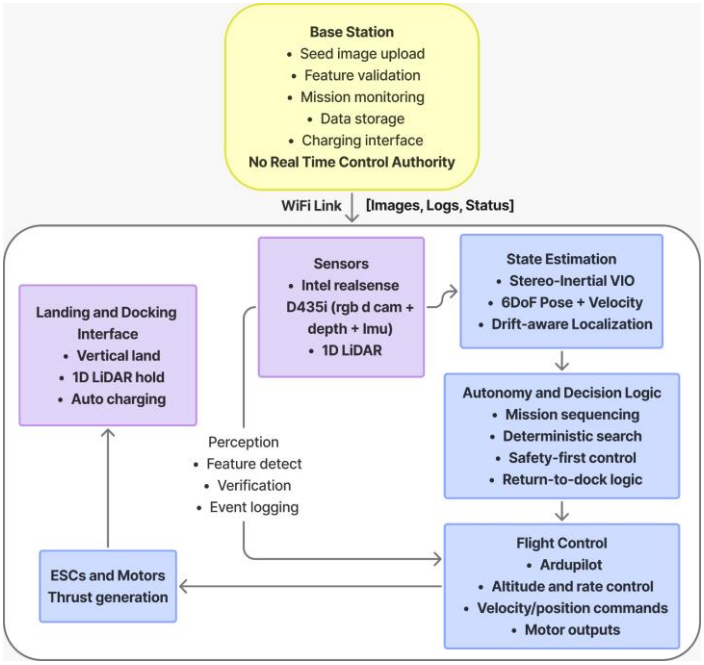


Figure 1: Block Diagram of Base Station and ASCEND UAV Integration

ASCEND ("Autonomous Surveyor Challenge for Exploration, Navigation and Dynamics") requires teams to build and develop a **micro rotorcraft UAV (<2 kg)** and a **base station**. The UAV must:

(i) **Seed** (ingest/collect provided seed feature images via the base station interface

(ii) **Search** an unknown indoor arena to locate instances of the seeded feature types

(iii) **Detect & Log local coordinates relative to the base station frame**

(iv) **Return** to the base station

(v) Transmit (images + coordinate log) to the base station then **Recharge,** enabling **repeat sorties**

Table 1: Comparative Analysis of Rotorcraft Configurations for Mission Suitability

Rotorcraft	Key Advantages	Key Limitations	Verdict
Quadrotor (X4)	Stable hover, simple & lightweight, long endurance, ideal for vision/landing/docking	No motor redundancy	**Optimal choice** reliable, efficient, competition-proven
Hexacopter (X6)	Higher lift capability	**≤ 2 kg** (clearer and standard-compliant)	Adds unnecessary system complexity
Single-Rotor Helicopter	Efficient hover	High mechanical complexity, poor automation for landing/docking	Operational risk is high
Tiltrotor / VTOL	Hover + fast forward flight	Unsuitable for docking and indoor search	Not aligned with mission requirements

The X4 quadrotor was selected for its **stable hover** and **precise low-speed control**, which are essential for stationary and slow-moving search operations in the ASCEND mission. Differential thrust control enables **accurate position holding** and **attitude stabilization**, supporting reliable visual-inertial odometry and reliable docking.

Considering the ASCEND mission constraints of sub-2 kg mass, indoor operation, autonomous docking, and repeat sorties, the X4 quadrotor provides the optimal balance between stability, control authority, endurance, and mechanical simplicity. Alternative configurations either exceed mass limits, reduce autonomy, or introduce unnecessary mechanical and control complexity without improving mission performance.

2.SYSTEM ARCHITECTURE

2.1 ONBOARD COMPUTING, SENSING, AND CONTROL STACK

Intel RGB-D Camera with IMU: Acts as the primary sensing unit, providing hardware-synchronized visual and inertial data for drift-aware localization and passive perception.
Downward 1D LiDAR: Supplies absolute altitude measurements during landing and docking, ensuring stable vertical control even under dust or low-contrast conditions.
Cube Flight Controller: Manages hard real-time stabilization, attitude control, and motor actuation, remaining independent of high-level autonomy and compute load.
Jetson Orin Nano Companion Computer: Runs all high-level autonomy including VINS-Fusion, perception and feature verification, mission FSM, and safety/health monitoring, while remaining isolated from flight-critical control loops

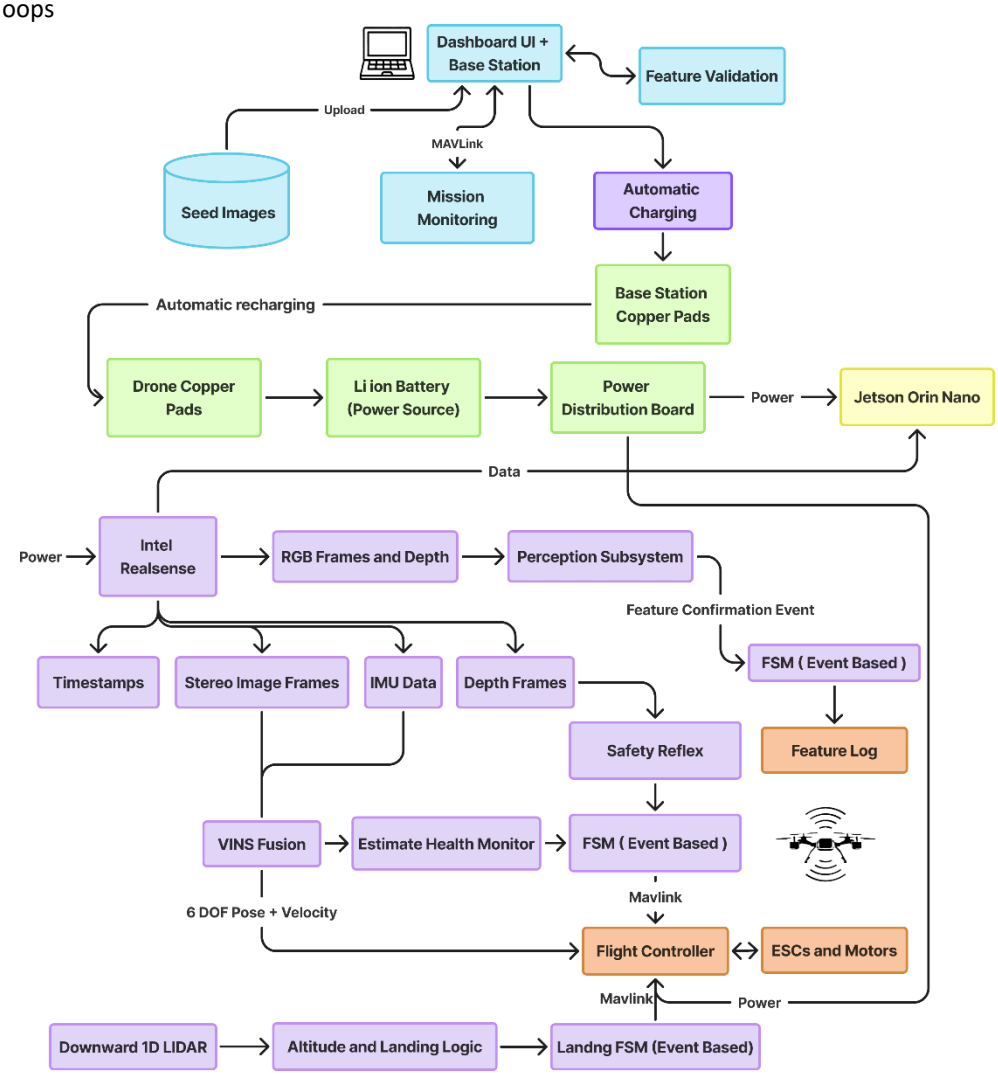


Figure 2:System Architecture of UAV

1. The entire autonomy stack is governed by a **single event-based finite-state machine (FSM)**.
2. **Event priority is strictly enforced in the following order: Safety → Health → Mission → Perception**
3. **Safety Reflex has absolute priority** and can override all mission actions at any time.
4. When a safety condition is active, **perception is muted** and no feature logging is allowed.
5. **The Health Monitor is advisory**: it raises degradation events but **never commands actuators directly**.

The design enforces a **clear top-down flow** — from sensing to estimation, decision-making, control, and actuation — while **safety and power management supervise all layers**. Each subsystem has a **single responsibility and explicit interface**, eliminating functional overlap and reducing integration risk. The subsystems:

1. Physical Sensing & Actuation Layer This layer forms the **hardware interface with the environment**. It provides raw, time-stamped sensor data and executes low-level motor commands, with **no embedded autonomy or decision logic**. Includes: Intel RGB-D camera with IMU, downward 1D LiDAR, ESCs, motors, battery and power distribution hardware.	**2. State Estimation Subsystem (Localization Core)** This subsystem provides real-time GNSS-denied localization, estimating the UAV's **6-DoF pose and velocity** using stereo-inertial visual odometry. It continuously monitors estimator health and drift. Outputs: **Metric pose and velocity → Flight Control** **Estimator confidence → Mission FSM & Safety**
3. Perception & Feature Extraction Subsystem (Passive, Event-Driven) Perception runs in **parallel to navigation** and is strictly decoupled from control. It detects and verifies seeded visual features using RGB-D data and reports events only, **never motion commands**. Outputs: **Feature confirmation events → Mission FSM** **Logged images & metadata → Data Handling**	**4. Mission Planning & Decision-Making Subsystem** All mission behavior is governed by a **single authoritative Finite State Machine (FSM).** The FSM executes deterministic mission sequencing and arbitrates between estimation of health, safety events, battery state, and mission progress. FSM States: INIT → READY → TAKEOFF → SEARCH → FEATURE_LOGGING → RETURN → LAND → DOCK → DATA_TRANSFER → RECHARGE → NEXT_SORTIE / END Outputs: High-level position and velocity setpoints → Flight Control
5. Safety, Health Monitoring & Fault Management (Supervisory Layer) This subsystem continuously supervises system integrity across all layers. It classifies faults as recoverable or critical and raises prioritized events to the FSM. Key Principle: Safety never commands motors directly; it retains veto authority but enforces all actions through the FSM to preserve deterministic behavior.	**6. Flight Control & Actuation Subsystem** This subsystem performs **low-level stabilization and control**, independent of perception and planning logic. It tracks position and velocity setpoints and converts them into motor commands. **Outputs:** Motor commands → ESCs & motors

7. Docking & Power Management Subsystem This subsystem enables **autonomous landing, docking, and persistent operation**. It manages vertical descent using range sensing, passive mechanical alignment, electrical contact detection, and autonomous battery charging. Power gating disables non-essential modules outside active mission phases.	**8. Communication & Middleware Subsystem** Provides structured data exchange between onboard modules using **ROS** and interfaces with the flight controller via **MAVLink (MAVROS)**. The architecture **explicitly excludes real-time teleoperation**, ensuring autonomy under communication loss
9. Data Handling & Time Synchronization Subsystem Ensures deterministic operation through **global time synchronization**, priority-based data handling, and **selective logging** of keyframes and mission summaries instead of raw streams.	**10. Ground Station & Base Infrastructure** (External to Onboard Autonomy) **Purpose** •Seed image upload before sortie •Mission monitoring (read-only) •Feature validation •Charging interface at the docking station

2.2 AUTONOMOUS BATTERY RECHARGING ARCHITECTURE

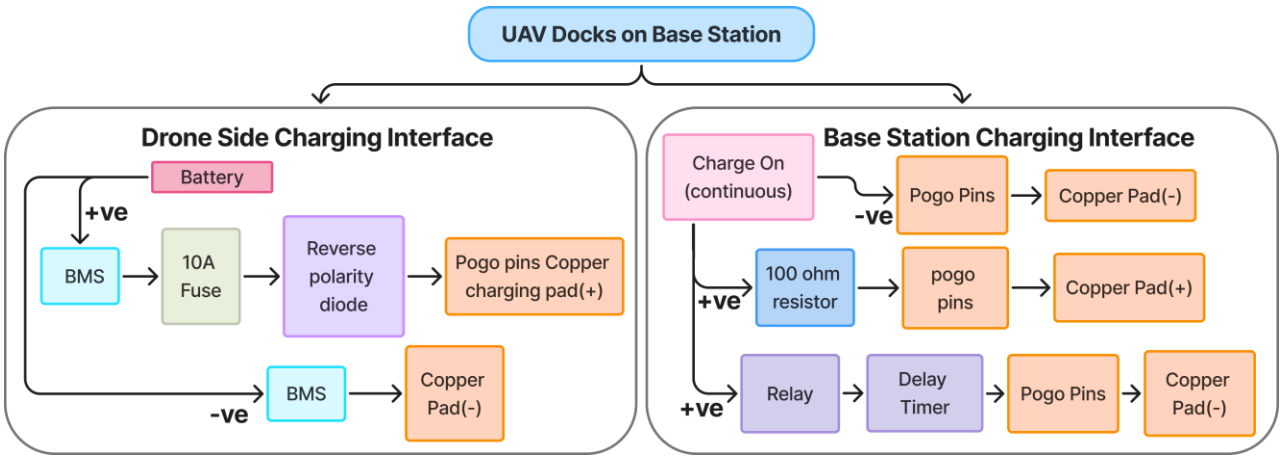


Figure 3: Autonomous Charging Flow

Charging Sequence (Autonomous, No Human Intervention):

1. Charger is powered ON continuously.
2. UAV lands and docks passively
3. Copper pads contact base-station pads
4. Pre-charge current flows through resistor (safe voltage equalization)
5. Delay relay activates after ~0.5 s
6. Full charging current is applied
7. BMS manages cell balancing and temperature
8. Charging terminates automatically at full capacity. This approach replicates **soft start charging behavior.**

3.IDENTIFICATION OF COMPONENTS WITH THEIR SPECIFICATIONS

3.1 PHYSICAL ARRANGEMENT OF ELEMENTS

The arrangement of the elements is carried out in such a manner that the center of gravity of the multirotor coincides with its geometric center. The detailed description and purpose of the individual components used in the UAV are presented in Table 2.

Table 2:UAV Component Description and Functional Purpose

No.	Component	Description	Purpose
1	**Frame Hub**	Quadcopter F450 frame with landing gear	Provides housing and mounting for all components
2	**Motor**	Sunny Sky X2212 KV980 brushless motor	Provides thrust and rotates propeller
3	**Propeller**	9×4.5-inch (9450) propeller	Generates lift by accelerating airflow
4	**Battery**	5600mAh 4S Li-ion battery	Supplies power to the UAS
5	**ESC**	SpeedyBee BLS 60A 4-in-1 ESC	Controls motor speed
6	**Flight Controller**	Pixhawk Cube Orange	Central control and stabilization system
7	**Power Module**	CubePilot Power Brick Mini	Power regulation and monitoring
8	**Companion Computer**	NVIDIA Jetson Orin Nano	Onboard computation and AI processing
9	**Depth Camera**	Intel RealSense D435i	Depth sensing and visual data acquisition
10	**LiDAR Sensor**	Benewake TF-Luna LiDAR (UART)	Distance and altitude measurement
11	**RC Receiver**	RadioMaster RP2 V2 ExpressLRS 2.4 GHz	RC communication and telemetry(failsafe)
12	**Telemetry Radio**	Holybro SiK Telemetry Radio V3 (433 MHz)	Long-range bidirectional data link (failsafe)
13	**Wi-Fi Adapter**	TP-Link Nano USB Wi-Fi 6 Adapter	Wireless data transmission

3.2 PROPULSION SYSTEM

3.2.1 MOTOR AND ESC SELECTION

The SunnySky X2212 (980 KV) motors were selected to achieve a **high thrust-to-weight ratio with low current draw**, which is critical for hover-dominant missions. Compared to higher-KV motors, this choice improves efficiency and reduces thermal stress during a sustained autonomous flight.
A 60A 4-in-1 ESC was selected to reduce wiring complexity and mass.

3.2.2 PROPELLER SELECTION

9 × 4.5-inch propellers are matched to the selected motors and 4S battery, balancing thrust, efficiency, vibration, and acoustic noise. Larger propellers increase efficiency but risk motor overheating, while smaller propellers increase RPM and vibration.

3.3 ELECTRICAL AND POWER ARCHITECTURE

3.3.1 BATTERY AND POWER DISTRIBUTION

The UAV is powered by a **4S 5600 mAh Li-ion battery**, selected for its high discharge capability and compatibility with the propulsion system. A dedicated power module supplies regulated voltage to avionics and provides real-time current and voltage monitoring to the flight controller.

3.3.2 AUTONOMOUS CHARGING INTERFACE

To enable autonomous recharging without human intervention, the UAV incorporates **passive copper charging pads** integrated into the landing gear.

Table 3:VINS-Fusion Selection Rationale

Aspect	VINS-Fusion (Stereo + IMU) – Selected	VINS-Mono	ORB-SLAM3 (Stereo VIO)	Optical Flow + IMU	LiDAR-Inertial Odometry
Scale Setup	Stereo+IMU	Mono+IMU	Stereo+IMU	Downward cam+IMU	LiDAR+IMU
Scale Accuracy	Absolute(metric)	Relative (drifts)	Absolute	High-dependent	Absolute
Hover Stability	High	Poor	Moderate	Moderate	High
Computational Load	Moderate	Low	High	Very low	Very high
Drift Behavior	Bounded, predictable	Accumulates	Loop closure dependent	Unbounded	Low
Verdict	Flight-safe & dock-safe	Unsafe for docking	Brittle under motion	No return-to-dock	High computation load

Justification:
VINS-Fusion was selected for its ability to solve scale geometrically, stable hover and precision landing. Also, predictable drift per sortie and leaves compute headroom for perception.

3.4 SENSOR SELECTION AND TRADE STUDY

Table 4:Depth and Visual Sensor Comparison

Feature	Intel RealSense D435i (Selected)	Luxonis OAK-D Lite	ZED 2i	Monocular RGB camera	Event Camera
Depth Method	Active stereo (IR)	Active stereo	Passive stereo	No true depth	Event based

IMU Synchronization	Hardware-synchronized	Limited	No onboard IMU sync	limited	good
Weight	~72 g	~61 g	~166g	~20g	~50g
ROS / VIO Maturity	High	Moderate	high	high	low
Integration Complexity	Low	Moderate	high	low	low
Verdict	Best overall balance	Sync & transparency risk	Too heavy	No metric	Risky

Selection: The Intel RealSense D435i provides **hardware-synchronized stereo and IMU data**, **mature ROS and VINS Fusion support, and reliable depth in low-texture indoor environments**. These characteristics make it well suited for visual–inertial odometry and safety monitoring in autonomy-critical missions.

Altitude & Landing Sensors — Why 1D LiDAR?

1. Direct metric altitude
2. High update rate
3. Dust-robust
4. Independent landing verification
5. Simple interface to FSM

Figure 4: RealSense™ Depth Camera D435i

Figure 5: Benewake TF-LUNA Micro LiDAR

Role: Precision landing, Safety reflex and Dock alignment tolerance

Table 5:Comparison between GNSS-denied navigation and Stereo-Inertial VIO

Requirement	Alternatives Considered	Chosen Approach	Reason
GNSS-denied navigation	Optical flow, LiDAR SLAM	**Stereo-Inertial VIO**	Only option with metric scale + stability
Depth on low-texture terrain	Passive stereo, monocular	**Active stereo (RealSense)**	Works on sand & uniform floors
Precision landing	Vision-only, barometer	**VIO + 1D LiDAR**	Redundant, dust-robust
Integration reliability	Custom sensors	**RealSense + ROS**	Proven ecosystem
Competition robustness	Complex planners	**Deterministic FSM + VINS**	Predictable, repeatable

4. ASCEND REALIZATION PLAN

At the realization phase, the ASCEND system transformed the conceptual projection into the full-scale realizable components. The team came up with a production-oriented realization plan that included hardware source, fabrication, additive manufacturing, and software implementation. Every component was classified according to where it was realized as well as being checked against performance, weight, and integration criteria. This also provided a traceable and systematic route between design purpose and functional deployment.

4.1 HARDWARE REALIZATION PLAN

The next table shows the specific realization plan of the ASCEND system hardware, i.e. the switching of the design completion step to the physical implementation step. All of them are divided by the type of their realization - off the shelf, in-house production or additive manufacturing and each of them is supplied with the main specifications and quantity information. Such a systematic process guarantees the manufacturability, procurement preparedness, and planned integration at the stage of realization.

Table 6:List of Components

Sr No.	Component Category	Finalized Component	Specifications	Source	Quantity
1	Quadcopter Frame	F450 quad frame	450mm Wheelbase Composite structure	Of the shelf	1
2	Brushless motors	SunnySky X2212 980KV	980 KV rating 3S–4S compatible	Of the shelf	4
3	60A ESC	SpeedyBee BLS 60A 4-in-1 ESC	60 A continuous current BLHeli-S firmware	Of the shelf	1
4	Battery	4s 5600 mAh Li-ion	Nominal voltage: 14.8V Capacity: 5600 mAh Li-ion	Of the shelf	1
5	Power Module	Cube Mini power brick	Voltage & current sensing 5 V regulated output	Of the shelf	1
6	Flight Controller	Cube Pilot +	Triple-redundant IMU ArduPilot compatible	Of the shelf	1
7	Companion Computer (On-Board)	Nvidia Jetson Nano – Orin	Quad-core ARM CPU CUDA-enabled GPU	Of the shelf	1
8	VIO – vision camera	Intel RealSense D435i	Stereo depth sensing Integrated IMU	Of the shelf	1
11	Battery Management System	10-15A	Over-current protection Cell balancing support	Of the shelf	1
12	Wifi Adapter	TP-Link Nano USB Wi-Fi 6 Adapter	2.4 GHz Wi-Fi USB interface	Of the shelf	1

13	Copper Power Plate	Thickness 1-2mm	1–2 mm thickness Low-resistance distribution	In-house Fabrication	1
14	Docking station	Wood	Wooden structure Ground charging support	In-house Developed	1

4.1.1 HARDWARE REALIZATION OF IN-HOUSE FABRICATED COMPONENTS

4.1.1.1 DOCKING STATION

The docking bay has been customized and made in-house as a special takeoff, landing, and charging point of the UAV. The main goal of the docking bay is to facilitate the repeatable autonomous landing and at the same time provide a stable electrical charging platform. First, the traditional pads were used, but there was no provision of charging and positional repeatability. Thus, a specialized docking bay was created that would have controls with mechanical guidance and built-in electrical contacts.

The docking bay is a structure made from wooden panels which are chosen because of the ease of machining, electrical insulation, structural stability, and low cost. The docking bay is fitted with copper charging pads inlaid into the docking bay surface and insulated electrically between the structure and the pads. These pads reserve a continuous potential difference that is fed by an external power source.

4.1.1.2 ONBOARD CHARGING INTERFACE

To provide in-flight battery charging and protection, an off-the-should 4S Li-ion Battery Management System (BMS) has been chosen and developed in-house. The chosen BMS offers cell balancing, over-voltage, under-voltage, over-current, and short-circuit protection, which is why it is applicable to autonomous charging activities.

The in-house-made copper contact pads are installed on the landing gear of the UAV, and electrical connections to its charging input terminals of the BMS are made with insulated, flexible conductors. Non-conductive spacers and heat-shrink insulation are used to give the frame electrical isolation. When the UAV meets the docking bay, the copper pads of the landing gear form connection with the copper pads on the ground, thereby forming the charging circuit. This enables the flow of current into the BMS that controls the process of charging and safely supplies power to the battery pack. The BMS automatically stops charging when the battery has reached the maximum permissible voltage, and in this way, unattended operation is safe.

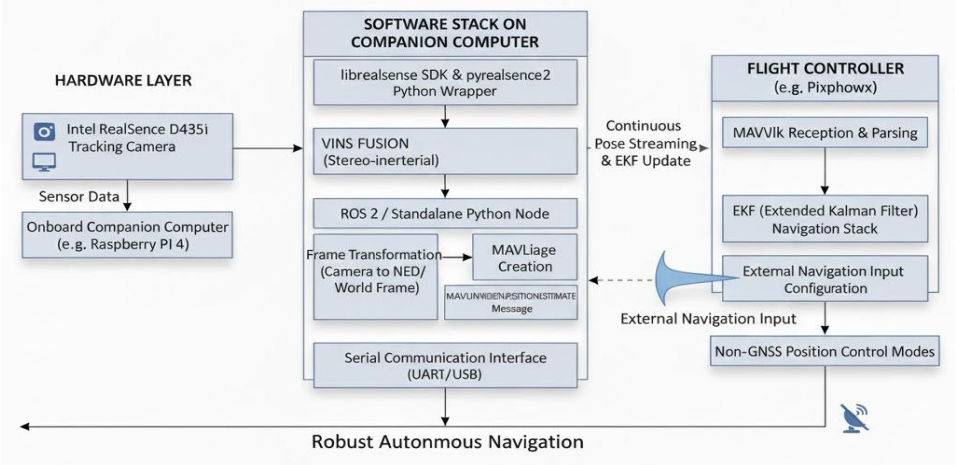
4.2 SOFTWARE REALIZATION

The realization of the ASCEND software stack is designed to enable fully autonomous navigation, exploration, target detection, logging, docking, data transfer, and repeated sortie execution **without any external navigation aids or human intervention**, in compliance with IRoC-U 2026 requirements.

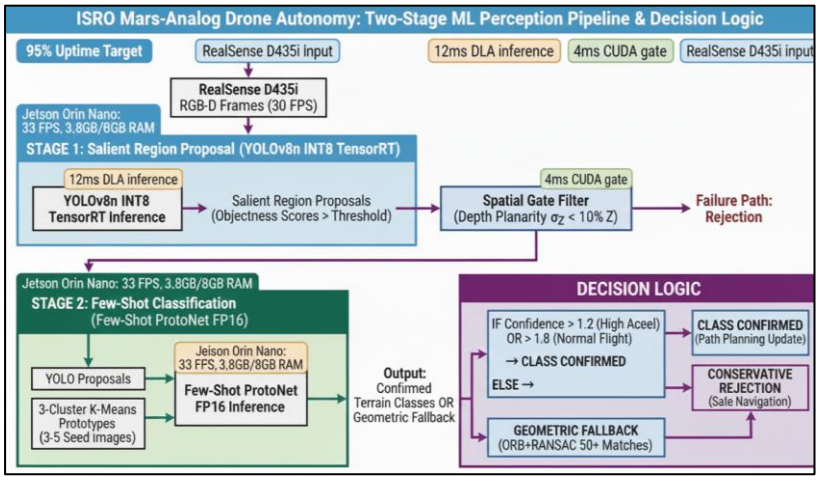
4.2.1 VISUAL – INITIAL ODOMETRY (VIO)

The subsystem of Visual-Inertial Odometry (VIO) is implemented with the help of an Intel RealSense tracking camera which can be interconnected with an on-board companion computer. Confidence data on pose and tracking are captured with librealsense SDK, and the information is processed with a special ROS 2 / Python

node where coordinate frame transformation is used to position the camera data within the vehicle navigation frame. The resulting transformed pose estimates are transmitted via MAVLink



4.2.2 OBJECT DETECTION, TARGET LOCALIZATION, AND LOGGING



During the flight, an object detection node will be running simultaneously with the VIO and navigation modules. The detection pipeline is startled as soon as the drone takes off and analyzes image streams provided by the onboard camera in real time. An object detector model is based on YOLO and is trained on the seed images when looking at the instances of target features in the arena.

When a valid target instance has been detected the detection node reads the associated VIO pose estimate and depth data and attempts to triangulate the relative three-dimensional coordinates of the detected target feature relative to the base-station reference frame. The detection location, coordinates, time of detection, and verification image are recorded with each detection. This data is stored in a structured format oboard, and it's traceable, and one can easily use it. The logging and the detection procedure should be used in parallel with the navigation and not disrupt the exploration path.

4.2.3 MISSION EXECUTION, DOCKING, AND REPEATED SORTIES

The exploration of the arena is conducted with the help of a lawnmower (boustrophedon) coverage strategy. There are waypoints to the exploration pattern which are created before the sortie and monitored with the help of constant VIO feedback. The mission manager node follows the progress and detection of occurrences of the exploration process during flight. Once a sortie is complete or the mission conditions are met, the drone finds its way back to the docking station independently using estimated positions locally. The system is capable of accuracy landing without human intervention after which the on-board software triggers the transfer of data of recorded images and coordinates to the ground station. The ground station compares the images that are transferred to the seeded reference images, as well as beginning battery charging. After validation and charging, the mission manager plans the next sortie which enables the drone to be redeployed autonomously. This software implementation in its closed-loop form can be used to perform multiple autonomous sorties in the

form of exploration, detection, landing, data transfer, and recharging, which is entirely fulfilling the operational goals of the IRoC-U 2026 challenge.

Software Layer	Method / Node	Role in Autonomous Mission	Technical Justification
Perception Layer	**YOLOv8-Nano**	Real-time region proposal (ROI generation)	Lightweight single-shot detector used only to narrow candidate regions, reducing downstream computation
Perception Layer	**ProtoNet (Few-Shot Classifier)**	Seed-image–based feature verification	Few-shot learning enables robust classification from limited reference images without retraining
Localization Layer (≈30 Hz)	**VINS-Fusion**	Primary 6-DoF visual–inertial odometry	Tightly coupled IMU pre-integration and stereo reprojection provide metric, drift-bounded localization
Decision Layer (≈10 Hz)	**Boustrophedon Coverage Planner**	Deterministic area coverage planning	Structured sweep-based planner provides systematic coverage with minimal overlap
Control Layer (50–100 Hz)	**Velocity / Position Controller (ROS-based)**	Trajectory tracking and local obstacle response	Decoupled control logic ensures smooth motion while respecting UAV dynamic constraints
Control Interface (100 Hz)	**MAVLink Bridge (Cube Orange)**	Command and telemetry interface to flight controller	Standardized high-rate protocol (921600 baud) ensures reliable offboard control
Supervisory Layer (≈30 Hz)	**Fail-Safe Monitor**	Fault detection and recovery escalation	Continuous monitoring enables tiered responses (hover, RTB, land) before instability
Data Layer	**Structured Telemetry Logger (JSON)**	Mission data persistence and GCS synchronization	Lightweight logging ensures data integrity during autonomous transfer

5. TESTING PLAN

The system is validated through a structured, multi-layer testing plan.

5.1 SUBSYSTEM-LEVEL VALIDATION: NAVIGATION & PERCEPTION

This phase validates the core intelligence of ASCEND, focusing on GNSS-denied localization, perception of robustness, and estimator reliability—key challenges in Mars-analog environments.

Table 9:Autonomous Mission Validation and Robustness Test Matrix

Test Category	Objective	Environment	Success Criteria
End-to-End Mission	Validate full autonomous mission execution	Mars analog indoor arena	Mission completed without human intervention

Autonomy Validation	Confirm decision-ma king without GCS	Comm-degraded	UAV maintains autonomous operations
Fault Tolerance	Validate system resilience	Fault-injected conditions	Safe recovery or controlled abort
Mission Repeatability	Verify consistency across sorties	ROS bag replay	Minimal variance in mission outcomes

Justification for Test Selection:

- **Drift Accumulation (MSN-02):** This test validates the *per-sortie reference reset* strategy by confirming that visual–inertial drift over a single sortie remains within the mechanical capture tolerance of the docking interface.
- **Comm-Denied Decision Making (LOG-01):** It verifies that the finite state machine (FSM) can manage all mission transitions exclusively using onboard logic without reliance on real-time ground station.
- **Endurance Consistency (RPT-01):** Verifies long-duration stability of the software stack, ensuring memory usage, thermal behavior, and estimator performance remain consistent across repeated sorties.

Table 10:Navigation Subsystem Validation and Failure Characterization Matrix

Test ID	Subsystem	Test Description	Test Method	Evaluation Parameters	Expected Outcomes
NAV-01	RealSense D435i	Camera-IMU time sync	Static & motion test	Sync error	<5 ms offset
NAV-02	VIO	Low-texture tracking	Sand/monochrome terrain	Feature loss	Stable pose estimation
NAV-03	VINS Fusion	Drift accumulation	100 m closed loop	Drift (cm/m)	<= 1% drift
NAV-04	VIO Recovery	Occlusion recovery	Temporary Lens obstruction	Recovery time	< 2 s
NAV-05	IMU	Bias stability	Extended hover	Bias drift	Within estimator bounds
NAV-06	VINS Failure	IMU saturation	Aggressive maneuvers	Estimator reset	Controlled fallback

These tests verify estimator robustness against texture loss, motion blur, occlusion, and inertial stress conditions typical of planetary analog terrains.

5.2. SOFTWARE ARCHITECTURE AND COMMUNICATION INTEGRITY

This phase validates the **middleware and runtime behavior** of the autonomy stack, ensuring deterministic operation under load and during partial failures.

Table 11:ROS Middleware Reliability and Runtime Stress Test Matrix

Test ID	Subsystem	Test Description	Method	Expected Outcomes
ROS-01	Topic Transport	Message latency	rosbag + delay	No command loss
ROS-02	TF Frames	Frame consistency	TF tree audit	Valid transform tree
ROS-03	Node Failure	Runtime node termination	Fault injection	Safe continuation
ROS-04	CPU Load	Long mission stress	60+ min run	Stable CPU usage
ROS-05	Memory Stability	Endurance memory test	Valgrind	No critical leaks

5.3 MECHANICAL, DOCKING AND PERCEPTION VALIDATION

This phase validates perception of robustness and mechanical reliability required for autonomous docking and persistent operation.

Table 12:Perception Robustness Validation Matrix

Test ID	Test Case	Condition	Metric	Expected Outcomes
PR-01	Feature detection	Normal lighting	Detection accuracy	>= 90%
PR-02	Dust interference	Artificial dust plume	Precision	>= 85%
PR-03	Partial occlusion	30–50% visual occlusion	Recall	>= 80%
PR-04	Runtime performance	Jetson Orin Nano	Stable frame rates	>= 30 FPS

Table 13:Autonomous Docking and Landing Validation Matrix

Test Case	Condition	Metric	Acceptance Criteria
Dock detection	Normal lighting	Detection accuracy	>= 90%
Precision landing	Normal Conditions	Position error	<= 3cm
Dust landing	Dust plume	Stability	No oscillation
Misalignment Handling	Offset docking	Safety response	Abort triggered

These tests confirm reliable feature recognition, stable vertical control, and correct execution of abort logic during docking failures.

6. BRIEF OF SYSTEM SPECIFICATION

The system specification of ascend defines the platforms size, power, sensing, computation, communication, and autonomy capabilities required to execute the mission reliably and safely.

Table 14: UAV Specifications

Sr no.	Description	Specification	
1	**Overall Mass (AUW)**	**1732.5gm**	
2	Overall Dimensions	Size	678*678*240(mm)
		Rotor span	683mm
3	Propulsion Configuration	Vehicle Type	Rotorcraft Micro-UAV
		Configuration	Quadcopter
		Propellers	4 propellers, each 9 inches in size
4	Power System & Requirements	Battery config.	4S Li-ion (14.8 V nominal)
		Energy capacity	82.9 Watt
		Recharge method	Autonomous contact-based docking station.
5	Endurance per Sortie	6-7min	
6	Maximum Flight Speed	2 m/s	
7	Navigation & Localization Capability	Method	Visual-Inertial Odometry (VIO)
		Sensors	RealSense RGB-D camera + IMU
		Output	Real-time 6-DoF pose & velocity
8	Autonomy & Control Architecture	Event based FSM, onboard decision making, Position/Velocity setpoints	
9	Perception & Task Capability	Intel RealSense RGB-D camera, depth-based ranging	
10	Safety & Fault Handling Mechanisms	Low Battery	Auto return-to-dock
		Signal Loss	Hover → return home → land
		Ground Safety	Geofence, arming lock, auto-disarm on landing
11	Landing & Docking Capability	Landing	Visual depth guided, precision descent
		Docking	Visual-guided landing on charging pad for auto recharge
12	Operational Concept (Sortie-Based Operation)	Autonomous sortie cycles, per sortie localization reset, repeatable missions	
13	Ground Station Role	Seed image upload, Feature validation, Automatic recharging	

7. OVERVIEW OF THE EMERGENCY RESPONSE SYSTEM

7.1 ANTICIPATED EMERGENCY SITUATIONS

Based on the task requirements the following emergency scenarios are anticipated:

1. **Localization Degradation**
2. **Obstacle Proximity / Collision Risk**
3. **Power-Related Emergencies**
4. **Communication Loss**
5. **Docking and Landing Anomalies**

7.2 EMERGENCY RESPONSE SYSTEM ARCHITECTURE

The emergency response system is implemented as an **event-driven safety layer tightly integrated with the mission finite state machine (FSM)**.

Table 15:Emergency Detection and Response Logic

Emergency Condition	Detection Mechanism	Response Action
VIO confidence degradation	Feature count, reprojection error, covariance growth	Abort exploration → Return to base → Controlled vertical landing
Obstacle proximity	Depth threshold breach (RGB-D / range data)	Immediate deceleration or yaw-away maneuver
Low battery	Power monitor threshold	Terminate search → Return-to-dock
Communication loss	Telemetry timeout	Continue mission autonomously or return based on FSM state
Camera failure	Frame timeout	Immediate controlled landing using inertial stabilization
Docking misalignment	Range + contact validation failure	Abort descent → retry or land nearby safely
Charging fault	Voltage/current anomaly	Disconnect charging path → safe idle state

All emergency actions are executed **on board**, without waiting for external approval.

7.3 FAIL-SAFE MODES IMPLEMENTED

The ASCEND system implements the following fail-safe behaviors:

Return-to-Base (RTB): Triggered by low battery, localization degradation, or mission timeout.
Safe Vertical Landing: Used when return is unsafe or localization confidence drops below recoverable limits.
Emergency Hover & Descent: Activated during transient perception or computation instability.
Docking Abort: Prevents mechanical or electrical damage in case of misalignment or unstable contact.

7.4 REDUNDANCY AND SAFETY MEASURES

While mass and complexity are constrained, functional redundancy is achieved through:
Sensor Diversity: Stereo vision + IMU + independent downward range sensor
Logical Redundancy: Safety reflexes operate independently of perception and exploration logic
State Isolation: Perception never influences flight control directly
Power Protection: Fuse, BMS, reverse-polarity protection, and staged charging current.

Table 16:Alternative Emergency Response Approaches Considered

Approach	Reason Not Selected
Manual override via GCS	Violates autonomous operation constraint

Estimator switching in flight	Introduces instability and unpredictable control
Aggressive obstacle avoidance planning	Increases computational load and failure modes
Redundant flight controllers	Weight and integration complexity

8. PROJECT MANAGEMENT

8.1 WORK BREAKDOWN STRUCTURE

The project activities can be divided into hardware development, software development, system integration, as well as testing and validation. The main role of each one of the categories is determined depending on technical expertise with secondary members backing up important subsystems in order to provide continuity and redundancy.

Table 17:Project Work Breakdown and Responsibility Assignment

No.	Task / Module	Key Responsibilities	Lead	Deadline
1	**Mission & Flight Ops**	Flight safety, Sortie execution, Piloting	Atharva Sharma	10-01-26
2	**Autonomy & FSM**	FSM design, mission sequencing, state transitions, autonomy logic.	Aditya Khairnar	02-02-26
3	**GNSS-Denied Nav**	VIO integration, relative localization, guidance, return-to-base.	Aarjav Jain	02-02-26
4	**Vision & Detection**	Feature extraction, detection logic, image capture, false-positive control.	Parth Mahajan	10-03-26
5	**Seed Image Management**	Autonomous intake, indexing, storage, and synchronization.	Jiya Patel	25-03-26
6	**Base St. Validation**	Seed-detection matching, validation pipeline, sortie verification.	Dhiren Jain	15-04-26
7	**Docking & Power**	Dock design, landing alignment, contact sensing, power transfer.	Siddhant Thakur	20-04-26
8	**Data & RF Comm**	UAV-based link, image/log transfer, bandwidth, & link optimization.	Mahak Sharma	24-04-26
9	**Integration & Fail-Safe**	End-to-end integration, emergency call-off, aborts, fault handling.	Ananya Mishra	25-05-26
10	**Testing & Compliance**	Test planning, performance validation, and DGCA/IRoC compliance.	Tanmay Porwal	31-01-26

8.2 STRATEGY FOR SCHEDULE MANAGEMENT

Table 18:RAD-Based Project Schedule and Phase-Wise Deliverables

RAD Cycle	Period	Primary Focus & Key Outputs
RAD-1 & Freeze-1	Dec 12 – Jan 12	Architecture formulation and Preliminary Round proposal submission.
RAD-2: Pre-Build	Jan 13 – Feb 02	Procurement planning, CAD finalization, and software stack initialization.
RAD-3: Integration	Feb 03 – Feb 28	Mechanical/electrical assembly and initial controlled flight testing.
RAD-4 & Freeze-2	Mar 01 – Apr 02	Sensor validation, PID tuning, and Qualification Round reporting.
RAD-5: Autonomy	Apr 21 – May 20	Development of autonomous navigation, docking, and charging.
RAD-6: Reliability	May 21 – Jun 15	Stress testing and multiple end-to-end autonomous mission executions.
Final Readiness	Jun 16 – Jul 19	Elimination Round lock, field rehearsals, and transport readiness.

Table 19:Subsystem-Wise Resource Allocation and Cost Breakdown (INR)

Subsystem	Component / Item	Quantity	Cost (INR)
Propulsion System	SunnySky X2212 KV980 Brushless Motors	4	6,456
	SpeedyBee BLHeli-S 60A 4-in-1 ESC	1	4,106
Power System	5200 mAh 4S1P Li-ion Battery (40–80C)	1	3,255
	CubePilot Power Brick Mini	1	2,659
	Wireless Charging Module	1	2,831
	Wiring Harness	1	100
Flight Control & Avionics	Hex Pixhawk Cube+ Flight Controller	1	38,555
	Holybro SiK Telemetry Radio V3 (433 MHz, 500 mW)	1	6,607
	RadioMaster RP2 V2 ExpressLRS Receiver	1	2,229
Computing & Communication	Jetson Orin Nano	1	26,443
	TP-Link Archer TX20U Nano (Wi-Fi 6 USB Adapter)	1	1,899
Sensors & Perception	Intel RealSense D435i Depth Camera	1	34,499
	Benewake TF-Luna LiDAR (8 m, UART)	1	2,110
Frame & Mechanical	F450 Quadcopter Frame with Landing Gear	1	798
	Copper Foil	1	100
	Miscellaneous (Mounts, Dampers,Wires,Screws)	1	250
Grand Total			**₹132,897**

9. NOVELTY IN THE OVERALL PROPOSAL

9.1 SORTIE-BASED, DRIFT-BOUNDED AUTONOMY (CORE NOVELTY)

Each mission sortie starts and ends at the same docking station.
Visual–Inertial Odometry (VIO) is **reset at the beginning of every sortie to origin (0,0,0)**
The docking station acts as a local reference of origin for each flight.
This approach **bounds drift by design**, rather than attempting to eliminate it. By ensuring that drift never exceeds the docking capture tolerance within a single sortie, ASCEND achieves reliable long-term operation without GPS, external markers, or global maps.
This reframing of drift—from a failure mode into a managed constraint—is a key originality of the system.

9.2 EVENT-BASED FSM WITH SAFETY VETO AUTHORITY

ASCEND is governed by a **single authoritative, event-driven Finite State Machine (FSM)** Key original aspects include:
Strict priority ordering: **Safety > Health > Mission > Perception**
Safety and health modules have **veto authority**, not control authority.
All actions are executed through the FSM, ensuring traceable and deterministic behavior.
This architecture is especially suited for competition and space-analog environments, where predictability and explainability are more valuable than aggressive optimization.

9.3 CONTROL: SENSOR HANDOVER PROTOCOL FOR PRECISION LANDING

The landing control logic enforces a **strict, non-overlapping sensor authority hierarchy**:
Above 1.0 m altitude: Full navigation and vertical control authority is assigned to VIO (Visual-Inertial Odometry)
Pre-Land Condition Triggered: VIO depth estimation is **hard disabled**
Below 1.0 m altitude: Vertical control authority is transferred exclusively to a **downward facing 1D LiDAR**
Benefits: This prevents instability that causes crashes in standard VIO drones, ensuring a smooth, vertical descent where dust, low texture, and lighting variations degrade camera performance.

9.4 PASSIVE, HARDWARE-ENFORCED AUTONOMOUS RECHARGING

ASCEND introduces a **passive, hardware-enforced charging architecture**:

Soft start charging using resistor + delayed relay.
No software-controlled charging switches.
Only copper pads are exposed, protected by fuse, diode, and BMS.

This ensures autonomous recharging **without sensors, actuators, or negotiation protocols**, reducing failure modes, and simplifying verification. The novelty lies in achieving autonomy through **electrical design**, not software complexity.

9.5 MOTION CONSTRAINTS TO LIMIT DRIFT GROWTH

During flight, the system enforces:
Constant altitude, Bounded forward velocity, Bounded yaw rate,
Smooth trajectories (no aggressive accelerations)